Lecture 23

# KdV as a Hamiltonian field

The Korteweg–de Vries equation uses one real field and a third spatial derivative. We describe it by a Hamiltonian and a bracket whose fixed operator is spatial differentiation. This bracket also has conserved quantities that generate zero motion.

**Objectives.** *We prove the Gardner bracket's Jacobi identity on a specified class of functionals. We derive KdV and its conservation laws, construct a travelling pulse and identify the obstruction to inverting its bracket operator.*

By the end of this lecture you will be able to:

- Compute the Gardner Hamiltonian vector field with the course signs.
- Prove conservation along a Schwartz solution and identify a Casimir.
- Derive the negative travelling pulse and its speed.

## 23.1 Fields and functional derivatives

Write $\mathcal{S} = \mathcal{S}(\mathbb{R}, \mathbb{R})$ for the real Schwartz space of Lecture 7. Its elements are the field data $u(x)$, and its seminorms are $\sup_x (1 + |x|)^N |\partial_x^j u(x)|$ for integers $N, j \geq 0$. The integral of the field has constant functional derivative. We therefore allow constants in the gradient space for this lecture.

**Definition 23.1** (Gradient space). The *gradient space* is the direct sum

$$\mathcal{B} = \mathbb{R} \oplus \mathcal{S} = \{a_0 + f : a_0 \in \mathbb{R},\ f \in \mathcal{S}\}.$$

Its topology is given by $|a_0|$ together with all Schwartz seminorms of $f$. For $a \in \mathcal{B}$ and $h \in \mathcal{S}$, use the pairing $\int_{\mathbb{R}} a(x)h(x)\,\mathrm{d}x$.

The decomposition is unique because a Schwartz function tends to zero at both ends. The pairing is finite and continuous: its absolute value is at most $|a_0|\|h\|_1 + \|f\|_2\|h\|_2$. The complete space $\mathcal{B}$ is a direct sum of $\mathbb{R}$ and the complete Schwartz space. The straight-segment calculus proved in Lecture 7 therefore applies to maps with this target as well.

**Definition 23.2** (Functionals for the Gardner bracket). A smooth functional $F : \mathcal{S} \to \mathbb{R}$ is *admissible here* if it has a smooth gradient map $F_u : \mathcal{S} \to \mathcal{B}$ satisfying

$$DF(u)h = \int_{\mathbb{R}} F_u[u]h\,\mathrm{d}x \qquad (h \in \mathcal{S}). \tag{23.1}$$

Smoothness means jointly continuous iterated directional derivatives, as in Lecture 7. Denote this class by $\mathcal{A}_{\mathrm{KdV}}$.

The gradient is unique by testing against compactly supported smooth $h$ and using the fundamental lemma. This argument applies even when the gradient is not in $L^2$. For $f \in \mathcal{S}$, the smeared field $u(f) = \int f u dx$ has gradient $f$. The functional $M[u] = \int u dx$ instead has gradient one in $\mathcal{B}$. It would lie outside the stricter Schwartz-gradient class of Lecture 7.

## 23.2 The Gardner bracket

Define the continuous linear operator $D_x : \mathcal{B} \to \mathcal{S}$ by $D_x a = \partial_x a$. If $a = a_0 + f$ and $b = b_0 + g$ lie in $\mathcal{B}$, then $ab \to a_0 b_0$ at both spatial ends. Integration by parts on $[-R, R]$ therefore gives, after taking the limit,

$$\int_{\mathbb{R}} a D_x b \, \mathrm{d}x = -\int_{\mathbb{R}} (D_x a) b \, \mathrm{d}x.$$

The two endpoint limits agree. They need not individually be zero.

**Definition 23.3** (Gardner bracket and Hamiltonian vector field)**.** For $F, G \in \mathcal{A}_{\mathrm{KdV}}$, the *Gardner bracket* is

$$\{F, G\}[u] = \int_{\mathbb{R}} F_u[u] D_x G_u[u] \, \mathrm{d}x. \tag{23.2}$$

Its *Hamiltonian vector field* is $X_F(u) = D_x F_u[u] \in \mathcal{S}$.

The integral pairs an element of $\mathcal{B}$ with an element of $\mathcal{S}$. It gives $X_G F = \{F, G\}$. In particular, the smeared bracket is $\{u(f), u(g)\} = \int f g' \mathrm{d}x$ for $f, g \in \mathcal{S}$.

**Lemma 23.4** (Closure and Jacobi)**.** *The class $\mathcal{A}_{\mathrm{KdV}}$ is closed under products and the Gardner bracket. The bracket is bilinear, antisymmetric, obeys the product rule and satisfies Jacobi.*

*Proof.* Let $H_F(u) = D(F_u)(u)$ denote the derivative of the gradient map, with $H_F(u) : \mathcal{S} \to \mathcal{B}$. Mixed scalar derivatives on $u + sh + tk$ commute, so

$$\int (H_F h) k \, \mathrm{d}x = \int h (H_F k) \, \mathrm{d}x \qquad (h, k \in \mathcal{S}).$$

Write $a = F_u$, $b = G_u$ and $a_K = K_u$, at the same field. Differentiation of the bracket, this symmetry and the skew integration by parts just proved give

$$\begin{aligned} D\{F, G\}(u)h &= \int \left[ (H_F h) D_x b + a D_x (H_G h) \right] \mathrm{d}x \\ &= \int h \left[ H_F D_x b - H_G D_x a \right] \mathrm{d}x. \end{aligned}$$

The new gradient is $H_F D_x b - H_G D_x a \in \mathcal{B}$. It is smooth: repeated differentiation gives finite sums of higher gradient derivatives evaluated on smooth Schwartz directions, with jointly continuous results in $\mathcal{B}$. This is the composition argument of Lecture 7, with the complete target $\mathcal{B}$. Products have gradient $FG_u + GF_u$, also smoothly in $\mathcal{B}$. These formulas prove closure and the product rule. Bilinearity and antisymmetry follow from the definition and integration by parts.

For Jacobi, substitution of this gradient into the cyclic sum gives

$$\begin{aligned} \int \big[ &(H_F D_x b) D_x a_K - (H_F D_x a_K) D_x b \\ &+ (H_G D_x a_K) D_x a - (H_G D_x a) D_x a_K \\ &+ (H_K D_x a) D_x b - (H_K D_x b) D_x a \big] \, \mathrm{d}x = 0. \end{aligned}$$

Each line cancels by Hessian symmetry. All the Hessian directions $D_x a$, $D_x b$ and $D_x a_K$ belong to $\mathcal{S}$. The argument never evaluates a Hessian on a constant direction outside the field's tangent space. ∎

## 23.3 Hamiltonian evolution and conserved integrals

The *mass*, *momentum* and *Hamiltonian* are the functionals

$$\begin{aligned} M[u] &= \int_{\mathbb{R}} u \,\mathrm{d}x, \qquad P[u] = \frac{1}{2}\int_{\mathbb{R}} u^2 \,\mathrm{d}x, \\ H[u] &= \int_{\mathbb{R}} \left(u^3 + \frac{1}{2}u_x^2\right) \mathrm{d}x. \end{aligned} \tag{23.3}$$

They are smooth polynomial functionals, with gradients $M_u = 1$, $P_u = u$ and $H_u = 3u^2 - u_{xx}$. For the last gradient, the derivative term on $[-R, R]$ obeys

$$\int_{-R}^{R} u_x h_x \,\mathrm{d}x = [u_x h]_{-R}^{R} - \int_{-R}^{R} u_{xx} h \,\mathrm{d}x.$$

Schwartz decay removes the displayed boundary term. Their gradient maps are continuous polynomial differential maps into $\mathcal{B}$, so all three belong to the stated admissible class.

**Proposition 23.5** (KdV as a Hamiltonian field)**.** *The Hamilton equation $u_t = X_H(u)$ for (23.3) is*

$$u_t = D_x \frac{\delta H}{\delta u} = 6uu_x - u_{xxx}, \qquad u_t - 6uu_x + u_{xxx} = 0. \tag{23.4}$$

*For every solution $u \in C^1(I; \mathcal{S})$ on an open interval $I$, all three functionals $M$, $P$ and $H$ are constant in time.*

*Proof.* Differentiating $H_u = 3u^2 - u_{xx}$ gives the stated equation. Along the assumed path, the chain rule gives $\dot F = \{F, H\}$. The mass derivative is $\int D_x H_u \mathrm{d}x = 0$ by decay. Antisymmetry gives $\dot H = \{H, H\} = 0$. For momentum, integration by parts gives

$$\{H, P\} = \int (3u^2 - u_{xx}) u_x \,\mathrm{d}x = \int \partial_x \left(u^3 - \frac{1}{2}u_x^2\right) \mathrm{d}x = 0.$$

Hence $\dot P = \{P, H\} = 0$ as well. ∎

The Hamiltonian need not be positive. These are identities along the stated Schwartz solutions. Lecture 24 cites the existence theory used in the later KdV lectures. The momentum's vector field is $X_P = u_x$, which generates the spatial translation $u(x) \mapsto u(x + \varepsilon)$.

## 23.4 The travelling pulse

Figure 23.1 shows a translated negative pulse at two times. Its shape and speed follow from the travelling-wave equation.

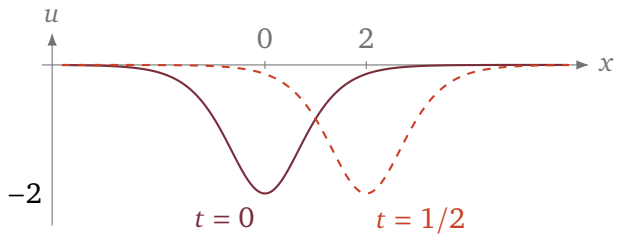



Figure 23.1: The pulse for $\kappa = 1$ and $x_0 = 0$, at $t = 0$ and $t = 1/2$. Its minimum moves from $x = 0$ to $x = 2$ while its profile stays fixed.

**Example 23.6** (A decaying travelling wave)**.** Find a negative decaying profile $u(t,x) = U(z)$ with $z = x - v_{\mathrm{tr}}t - x_0$ and $v_{\mathrm{tr}} > 0$. The resulting family, parametrised by $\kappa > 0$ and $x_0 \in \mathbb{R}$, is

$$u(t,x) = -2\kappa^2 \operatorname{sech}^2\big(\kappa(x - 4\kappa^2 t - x_0)\big). \tag{23.5}$$

To derive it, substitute into KdV and integrate once, using decay of $U$ and its derivatives:

$$U''' - v_{\mathrm{tr}}U' - 6UU' = 0, \qquad U'' - v_{\mathrm{tr}}U - 3U^2 = 0.$$

Multiplication by $U'$ and another integration give

$$(U')^2 = v_{\mathrm{tr}}U^2 + 2U^3.$$

At the negative minimum, $U' = 0$ and $U \neq 0$, hence $U = -v_{\mathrm{tr}}/2$. Place this minimum at $z = 0$. Set $w(z) = -2U(z)/v_{\mathrm{tr}}$. On $z > 0$, take the decreasing branch from $w(0) = 1$. Separation gives

$$(w')^2 = v_{\mathrm{tr}}w^2(1 - w), \qquad \operatorname{artanh}\sqrt{1-w} = \frac{\sqrt{v_{\mathrm{tr}}}}{2}z.$$

Thus $w = \operatorname{sech}^2(\sqrt{v_{\mathrm{tr}}}z/2)$. Its even extension gives (23.5), with $v_{\mathrm{tr}} = 4\kappa^2$. Direct substitution into the second-order equation verifies the extension at the minimum, where the separated first-order equation changes branch. The profile and all its derivatives decay exponentially, so its translations form a smooth Schwartz path.

The substitution $y = \tanh(\kappa z)$ makes its invariant integrals polynomial. It gives

$$M = -4\kappa, \qquad P = \frac{8}{3}\kappa^3, \qquad H = -\frac{32}{5}\kappa^5.$$

For example, $P = 2\kappa^3 \int_{-1}^{1}(1-y^2)\mathrm{d}y = 8\kappa^3/3$. These values are independent of the translated centre.

## 23.5 The Casimir and the derivative's domain

A *Casimir* is an admissible functional whose bracket with every admissible functional is zero. The mass is a Casimir because

$$\{M, F\} = \int_{\mathbb{R}} D_x F_u \,\mathrm{d}x = 0, \qquad X_M = D_x 1 = 0.$$

Its nonzero gradient lies in $\mathcal{B}$ and generates zero motion.

The domain matters when discussing invertibility. On $\mathcal{S}$, the derivative has zero kernel: a constant Schwartz function must be zero. Its range is precisely

$$\mathcal{S}_0 = \{f \in \mathcal{S} : \int f \,\mathrm{d}x = 0\}.$$

Every derivative has zero integral. Conversely, for $f \in \mathcal{S}_0$, the primitive $h(x) = \int_{-\infty}^{x} f(y)\mathrm{d}y$ is Schwartz. On the positive half-line use $h(x) = -\int_{x}^{\infty} f(y)\mathrm{d}y$ to estimate its rapid decay; the negative half-line uses the original formula. Its higher derivatives are derivatives of $f$. Thus $D_x : \mathcal{S} \to \mathcal{S}$ is injective but not onto. On $\mathcal{B}$, its kernel is the one-dimensional space of constants, and its range remains $\mathcal{S}_0$.

The bracket therefore has a Casimir and no inverse on the full gradient space. A finite-dimensional Liouville–Arnold theorem cannot be applied to this infinite-dimensional degenerate bracket. Lecture 27 uses different structures to organise its conserved quantities.

**Summary.** The Gardner bracket is $\{F, G\} = \int F_u D_x G_u$. Smooth gradients in $\mathbb{R} \oplus \mathcal{S}$ include the mass derivative one. Closure and Jacobi use Hessian symmetry on the genuine Schwartz directions $D_x F_u$. The Hamiltonian $\int (u^3 + u_x^2/2)$ gives KdV. Mass, momentum and energy are constant along its $C^1$ Schwartz solutions. The negative travelling pulse has speed $4\kappa^2$. The mass Casimir and the derivative's range prevent a global inverse for the bracket operator.

**Self-check.** Why is $M[u] = \int u$ admissible here although its gradient is not Schwartz? Why does it generate zero evolution?

**Answer.** Its gradient is one in $\mathcal{B} = \mathbb{R} \oplus \mathcal{S}$. The pairing with every Schwartz variation is finite. The operator $D_x$ kills this gradient, so $X_M = 0$ and $M$ is a Casimir.

**Self-check.** The Hessian of an admissible functional acts only on Schwartz tangent vectors. Why are all its arguments in the Jacobi proof allowed?

**Answer.** The arguments are derivatives of gradients, such as $D_x G_u$. Differentiation removes the constant part and leaves a Schwartz function. No Hessian is applied to the constant function one as a tangent direction.

**Self-check.** For $\kappa = 2$, give the travelling pulse's minimum and speed. What happens to its centre if $x_0$ increases?

**Answer.** The minimum is $-8$ and the speed is 16. Its centre is $x_0 + 16t$, so increasing $x_0$ shifts the profile to the right.

**Self-check.** Can $e^{-x^2}$ be the derivative of a Schwartz function? Compare the kernel of $D_x$ on $\mathcal{S}$ with its kernel on $\mathcal{B}$.

**Answer.** No. Its integral is $\sqrt{\pi} \neq 0$, whereas a Schwartz derivative has zero integral. The kernel on $\mathcal{S}$ is zero. The kernel on $\mathcal{B}$ consists of constants. These are different domain statements.